

COMPUTING APPLICATIONS CERTIFICATE

The Certificate in Computing Applications is a cross-disciplinary course of study in the Colleges of Liberal Arts and Sciences, Engineering, and Business. It is designed for undergraduates not already enrolled in majors in Computer Science, Software Engineering, Computer Engineering, or Cyber Security Engineering who wish to enhance their degree and employment possibilities by adding expertise in computing applications. The certificate program focuses on teaching students the essential skills required to develop and use computing applications in their subject domains. This certificate program will help students to be successful in today's workplace, ensuring they have the basic knowledge of programming languages and computer systems needed to enhance their employment opportunities.

The certificate program is offered jointly among the three colleges and their respective departments housing the majors of computer science, computer engineering, software engineering, and management information systems.

The program consists of 9 credit hours selected from the core and 12 credit hours in electives.

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Part I - Core 9-10 Credits

Students will complete either Sequence 1 or Sequence 2

SEQUENCE 1	9
	Credits

COMS 1060	Introduction to Web Programming	3
or COMS 1130	Introduction to Spreadsheets and Databases	

COMS 1270	Introduction to Computer Programming	3
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COMS 2520	Linux Operating System Essentials	3
or DS 2010	Introduction to Data Science	

SEQUENCE 2	10
	Credits

COMS 2270	Object-oriented Programming	4
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COMS 2280	Introduction to Data Structures	3
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COMS 2520	Linux Operating System Essentials	3
or DS 2010	Introduction to Data Science	

Part II - Electives 12 Credits

Designated Electives for the Certificate

Students will choose one elective course from the following list of courses and are responsible for satisfying the course pre-requisites of the elective selected.

- CPRE 2300 Cyber Security Fundamentals
- BCBIO 3220 Introduction to Bioinformatics and Computational Biology
- LA 2110 Digital Design Methods for Landscape Architecture
- CRP 2510 Fundamentals of Geographic Information Systems
- STAT 3201 Intermediate Statistical Concepts and Methods
- PSYCH 3010 Research Design and Methodology
- SOC 3020 Research Methods for the Social Sciences
- ECON 2070 Applied Economic Optimization
- Any course at 3000 level or above in Computer Science (COMS), Computer Engineering (CPRE), Management Information Systems (MIS), or Software Engineering (SE)

The remaining 9 credits of electives are intended to allow students to develop further computing applications expertise especially within their major. These 9 elective credits can be advanced courses in their major, courses on the above electives list, or any other coursework approved by the Computer Science Advisor on the "request for undergraduate certificate." At least 9 of the 12 elective credits must be in courses numbered 3000 or above and taken at Iowa State University.

The Computer Applications Certificate is an LAS Certificate and must meet University-wide policies for undergraduate certificates (<https://catalog.iastate.edu/academics/#degreeplanningtext>).